

# Lightning Finals

MNYMO Girls 2026

August 15th, 2026

# Rules

The top individuals will compete two-at-a-time in a single-elimination-style bracket tournament. For each match, a problem will be read and projected and the students will have 60 seconds to solve the problem. When a student has an answer ready, they may buzz in and answer and time is paused. You must wait for acknowledgment before answering.

# Rules (cont.)

If it is correct, they gain a point. If it is incorrect, time is resumed and the other student will have the remainder of the time to buzz in and answer. The first student to two points wins the match, and this process is repeated until a first-place winner is decided.

# Rules (cont.)

If there is no winner after three problems, the student with the higher score wins. Otherwise, the match will enter sudden death, where the next correct answer wins.

# Tiebreaker

Line  $\ell$  in the coordinate plane goes through the point  $(2, 3)$  and is perpendicular to the line  $y = \frac{1}{5}x + 10$ . What is the  $y$ -intercept of line  $\ell$ ?

**Answer: 13**





# Tiebreaker

What is the sum of the unique zeroes of the polynomial  $x^3 - 6x^2 + 9x - 4$ ?

**Answer: 5**



How many ways are there to pick four squares in the same row, column, diagonal, or  $2 \times 2$  square within a  $4 \times 4$  grid?

**Answer: 19**



A fair 6 sided die is rolled twice. What is the probability that the product of the numbers rolled by the dice are prime.



**Answer:  $1/6$**



What is the area of a triangle with side lengths  
5, 9, and 12?

**Answer:**  $4\sqrt{26}$



Sneha and Elizabeth are playing a game. They each receive the number 1, 2, or 3, chosen uniformly at random. The player with the larger number wins (no one wins if they tie). After seeing her own number, Sneha is given the choice to switch to one of the other two numbers (chosen randomly). Given that they can only see their own numbers and Sneha plays optimally, what is the probability that Sneha wins?

**Answer:**  $\frac{1}{2}$





What is the area of a regular octagon with a side length of 2?

**Answer:**  $8 + 8\sqrt{2}$



I am thinking of two numbers. One is the largest double-digit multiple of 7, and the other is the smallest two-digit multiple of 8. What is the sum of my two numbers?

**Answer: 114**



What is the perpendicular distance from the point  $(2, 4)$  to the line  $3x + 4y = 2$ ?

**Answer: 4**





If the expression  $\frac{16 \log_2(2^{16} \cdot 2^8)}{6}$  equals  $2^n$ , where  $n$  is an integer, what is  $n$ ?

**Answer: 6**



Angie is ordering at a boba shop that has 10 tea flavors and 5 toppings. Her order must include exactly 1 tea flavor, and she can choose to add either 0 or 1 topping. How many different unique orders can Angie place?

**Answer: 60**



Alice, Blice, and Clice are eating a pie. If Alice eats  $\frac{1}{5}$  of the pie, Blice eats  $\frac{2}{5}$  of the remaining pie, and Clice eats  $\frac{3}{5}$  of the remaining pie after Alice and Blice finish eating, how much pie would be left for their friend Dlice?



**Answer:**  $\frac{24}{125}$



Alice is buying a lottery ticket. The lottery ticket requires Alice to guess three digits from 0 to 9. If Alice guesses all three digits correctly in the correct position, she gets 100 dollars. If Alice guesses exactly two of the digits in the right position, she gets 10 dollars. And if Alice guesses exactly one digit correctly and it is in the right position, she gets 1 dollar. If Alice buys the ticket for 5 dollars, what are her expected winnings? Round to the nearest cent.

**Answer:  $-4.39$**



A goat is tied to the corner of a rectangular barnyard with dimensions  $6m$  by  $7m$ . If the goat's leash is  $10m$  long, what is the total area the goat is allowed to travel? Express your answer in terms of  $\pi$ .

**Answer:**  $\frac{325\pi}{4}$





What is  $1 + 4 + 9 + \dots + 576$ ?

**Answer: 4900**



How many distinct orderings of “minnesotan”  
are there?

**Answer: 604800**



A right triangle is situated on the coordinate plane with coordinates  $(0, 0)$ ,  $(4, 0)$ , and  $(0, 3)$ . If it is rotated about the x-axis by  $360^\circ$ , what is the surface area of the resulting figure? Express your answer in terms of  $\pi$

**Answer:**  $24\pi$





A deck contains an unlimited number of cards in each of three suits. What is the expected number of cards that must be drawn before at least one card from each suit has been drawn?

**Answer:**  $\frac{11}{2}$



How many ways are there to get from  $(0, 0)$  to  $(2, 3)$  on the coordinate plane by moving one unit up or right at a time and not passing through  $(1, 1)$ ?

**Answer: 4**



Triangle  $ABC$  has side lengths  $AB = 26$  and  $BC = 42$ . The altitude to  $BC$  has length 24.  
What is the length of  $AC$ ?



**Answer: 40**



How many distinct pairs  $(x, y)$  of positive integers are there such that  $xy = 2x + y + 4$ ?

**Answer: 4**



The prime factorization of  $20!$  can be expressed as  $2^a \cdot 3^b \cdot 5^c \cdot 7^d \cdot 11^f \cdot 13^g \cdot 17^h \cdot 19^j$ . Find  $a$ .

**Answer: 18**





Kaining has 5 balls of yarn, each of which measures 130 yards when unrolled. She wants to crochet penguins. Each penguin requires 60 yards. How many complete penguins can she make?

**Answer: 10**



Eric multiplies the ages of his 2 siblings and gets 45. If the sum of these 2 ages is divisible by 7, what is the age of the older sibling?

**Answer: 9**



Kaining, Elizabeth, and Michael each ate  $\frac{3}{5}$  of a sandwich from Sneha's Sandwich Shop. How many sandwiches did they eat in total?

**Answer:**  $\frac{9}{5}$





Find the positive difference between  $2^{(3^2)}$  and  $2^{(2^3)}$ .

**Answer: 256**